

Homework 4 - Stat 495 - Fall 2025

YourNameGoesHere

2025-10-17

Homework 4 - Due Monday, Nov. 3rd by midnight to Gradescope

Practicing Academic Integrity

If you worked with others or used resources outside of provided course material, such as Generative AI (anything besides our textbook, course materials in the repo, labs, R help menu) to help complete this assignment, please acknowledge them below using a bulleted list. Remember that according to our class framework, if you engage with Generative AI tools, the tool and date of use must be listed here, and then a complete record of the chat history must be appended to the assignment.

I acknowledge the following individuals with whom I worked on this assignment:

Name(s) and how they assisted you

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I used the following sources to help complete this assignment:

Source(s) and where used in your submission

-

PROBLEMS TO TURN IN: Add 1, Add 2, and Add 3

The first two problems use the bbb2 data set from the QSARdata package for Chapter 17 and 18 material. The final problem covers concepts from Chapter 15.

For the applied problems, the response variable of interest is bbb2_Class, which can be found in the bbb2_Outcome data set. We have joined the outcome variable to the QuickProp data set that we want to focus on below. You can read the associated help file in R to learn more about the data set. You saw part of this data set on the exam.

```
data(bbb2)
#?bbb2 # for variable reference and information
mybbb2 <- left_join(bbb2_QuickProp, bbb2_Outcome) %>% select(- Molecule)
```

Joining with `by = join\_by(Molecule)`

```
tally(~ bbb2_Class, data = mybbb2)
```

```
bbb2_Class
Crosses DoesNot
      45      35
```

Our goal for the applied problems below is to use the recent methods from class (Chapter 17 and 18) to predict the response variable, whether each compound “crosses” the blood-brain barrier or “does not” cross. You should use the *mybbb2* data set going forward. Note that there are 51 variables at the moment, and the last variable “Class” is the target, but if you open the data set to view, it will only show 50 variables by default. Class is there, but you have to use the arrows to see it.

```
# We loaded a lot of data sets we don't need anymore
# remove them to clean up your workspace
remove(bbb2_AtomPair, bbb2_Daylight_FP, bbb2_Dragon, bbb2_Lcalc, bbb2_moe2D,
       bbb2_moe2D_FP, bbb2_moe3D, bbb2_Outcome, bbb2_PipelinePilot_FP,
       bbb2_QuickProp)
```

To avoid issues with reproducibility, you should set a seed in EACH chunk below where you do a random process, whether that is setting up the train/test split or fitting a model that has some random process involved.

Note that because some of these methods can take a long time to tune, we will NOT be tuning for them. Instead, you will run them using their original packages/functions *ranger*, *gbm*, and

nnet with the inputs specified. This is to introduce you to those different inputs and what they are for/what they do. If you end up applying these models as some part of the final project, be sure you tune the models appropriately! You should use the help function for information about the functions if you are having issues running the models as a starting point.

Additional Problem 1

Your task for this problem is to do some setup, then fit the models described in parts c, d and e, and then compare them in part f. Use all available predictor variables (except what is removed in part a), with no re-expressions. The same training/test split will be used in Additional Problem 2 as well (i.e. you only make this once).

part a. One variable in the data set, QikProp_.amidine causes issues with some of these methods. We will remove it here, but can you see why it is problematic? Explain why this variable is not very useful for this analysis.

SOLUTION:

```
# run once you are ready to remove the variable to proceed
# this variable is not used anywhere below, so overwrite the data set
mybbb2 <- select(mybbb2, -QikProp_.amidine)
```

part b. Create an appropriate training/test split from mybbb2 with a ratio of 70/30 to use throughout the problems. As always, be sure your split is reproducible.

SOLUTION:

part c. Create an appropriate model to predict Class with the training set using bagging with 1000 trees and otherwise with default settings for tuning parameter values.

SOLUTION:

part d. Create an appropriate model to predict Class with the training set using a random forest with 1000 trees and otherwise with default settings for tuning parameter values.

SOLUTION:

part e. Create an appropriate model to predict Class with the training set using boosting with 500 trees and otherwise with default settings for tuning parameter values.

SOLUTION:

part f. Parts c, d, and e only required you to fit models. Now we want to compare their performance. Use an appropriate measure to compare the three models in terms of their model performance based on the test set. Be sure your choice of measure is clear. Summarize your findings. Then discuss which model you would choose for predicting Class. Explain your choice. (You can use additional assessment items, but must have the test set comparison.)

SOLUTION:

Additional Problem 2

Your task for this problem is to fit the models described in parts a, b, and c, and then compare them in part d. Use all available predictor variables, with no re-expressions. Use the same training/test data as above. Finally, in part e, you'll try to do better with a model of your own.

(Remember, these tuning parameter values were arbitrary choices I made so you can see the inputs and such to the models. They are not tuned for optimal performance!)

part a. Create an appropriate model to predict Class with the training set using a neural net with a single hidden layer of 12 nodes, and otherwise with default settings for tuning parameter values.

SOLUTION:

part b. Create an appropriate model to predict Class with the training set using a neural net with a single hidden layer of 12 nodes, and a decay parameter of $6e-3$, and otherwise with default settings for tuning parameter values.

SOLUTION:

part c. Create an appropriate model to predict Class with the training set using a neural net with a single hidden layer of 12 nodes, a decay parameter of $6e-3$, and a value for maxit that allows for convergence, and otherwise with default settings for tuning parameter values. (Ask if you aren't sure what the convergence idea is about!)

SOLUTION:

part d. Parts a, b, and c only required you to fit models. Now we want to compare their performance. Use an appropriate measure to compare the three models in terms of their model performance based on the test set. Be sure your choice of measure is clear. Summarize your findings. Then discuss which model you would choose for predicting Class. Explain your choice.

SOLUTION:

part e. Fit a new neural net model with parameters chosen by you (you can even explore options we didn't use above), but must set the size of the hidden layer and ensure your model reaches convergence. What parameter values did you use? Does your model do better than your final model in part d.?

SOLUTION:

Additional Problem 3

In a few sentences, explain to someone currently taking STAT 135 (so their level of background knowledge is intro stats, just having seen hypothesis testing for the first time), what p-value adjustment is all about and why it's an important concept to understand.

SOLUTION: